

FairML

Project Study Guide

Understanding the group–individual fairness trade-off study
from first principles

Overview • Terminology • Metrics • Methodology • Results •
Paper Preparation

How to read this document. Every section starts with a plain-language explanation and then gives the technical version. Blue boxes are worked numerical examples you can verify by hand. Amber boxes flag things that are wrong, misleading, or easy to misread in the current project. Green boxes are the ideas worth memorising.

Everything here is grounded in the actual code and result files. Where a value has no accepted standard, this guide says so explicitly rather than inventing a threshold.

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1 Project Overview

1.1 What problem are we trying to solve?

When a machine-learning model makes decisions about people — will this defendant re-offend? should this person get a loan? does this person earn more than \$50,000? — it can be unfair in **two different ways**, and those two ways are not the same thing.

Kind of fairness	The question it asks	Example complaint
Group fairness	Are demographic groups treated equally <i>on average</i> ?	“Black defendants are flagged high-risk far more often than white defendants.”
Individual fairness	Are the model’s mistakes spread evenly across <i>individuals</i> , or do a few people absorb all the harm?	“Overall the groups look balanced, but a small set of people get wrecked by the errors.”

The uncomfortable part: **fixing one can break the other**. Forcing every group to have the same approval rate means you must approve some people you would otherwise reject, and reject some you would otherwise approve. Those forced flips are extra errors, and they land on specific individuals.

There is a well-known body of theory saying certain fairness definitions are mathematically incompatible. What is much thinner in the literature is **empirical measurement**: if you actually build a model and turn a dial from “optimise group fairness” toward “optimise individual fairness,” what happens to each metric, on real data, with real error bars?

1.2 What is the main objective?

The one-sentence framing

The composite loss is the **instrument**; the trade-off map is the **contribution**.

We are *not* claiming “we invented a better fairness algorithm.” We are claiming: “we built a dial, turned it across three datasets $\times$ two model types $\times$ ten random seeds, and here is the empirical map of what the trade-off actually looks like.”

Three research questions drive everything:

- RQ1** Is there a *measurable* tension between group fairness (DPD/EOD) and individual fairness (GE) when each is optimised directly?
- RQ2** Is the tension consistent across datasets (race/COMPAS, sex/German, sex/Adult) and model classes (logistic regression, MLP)?
- RQ3** Where does our composite-loss frontier sit relative to established pre-, in-, and post-processing fairness methods?

1.3 Why does this problem matter?

- Real deployments must pick one.** A bank or court system implementing a “fairness requirement” has to choose a definition. If the definitions trade off against each other, that choice has a real cost, and practitioners deserve to know its size.
- Regulation is definition-blind.** Rules say “don’t discriminate,” not “keep demographic parity difference below 0.05.” Empirical trade-off maps are what translate a legal requirement into an engineering target.
- Impossibility theorems say “you can’t have both,” not “how much does each cost.”** Ours is a magnitude study.

1.4 What is the overall approach?

Build a single loss function with two knobs.

- α controls how much you care about *accuracy versus fairness at all*.
- β controls, *within* the fairness part, how much is group fairness versus individual fairness.

Then sweep a 7×7 grid of (α, β) , training a model at every grid point, on three datasets, with two architectures, repeated over ten random seeds — about **2,940 trained models** — and measure six metrics on each. Compare against six established baselines. Test significance with a paired non-parametric test across seeds.

2 Important Terms and Concepts

2.1 The basic setup

Binary classification

The model outputs one number per person, and you threshold it at 0.5 to get a yes/no decision. All three datasets in this project are binary.

Sensitive attribute (also: protected attribute)

The demographic variable that fairness is measured *with respect to*. In this project: `race` on COMPAS, `Sex` on German, `sex` on Adult.

Positive class

The outcome coded as 1. We deliberately made this the *adverse* outcome on COMPAS and German so the datasets align: COMPAS `two_year_recid` = re-offended; German `risk` = 1 means **bad** credit. Adult breaks the pattern — there 1 = income > \$50K, a *good* outcome.

Logit

The raw pre-sigmoid score. Our models return logits; the sigmoid is applied inside the loss. (This fixed a double-sigmoid bug in the legacy code.)

Sigmoid

Squashes a logit into a probability: $\sigma(z) = 1/(1 + e^{-z})$.

BCE (Binary Cross-Entropy)

The standard accuracy loss for binary classification. Penalises confident wrong answers heavily.

Design choice reviewers will ask about

The sensitive attribute is **left in the feature matrix** and fed to the model. This is a legitimate choice (“fairness through awareness” — hiding the attribute doesn’t help because proxies exist), but state it deliberately in the paper rather than leaving it implicit.

2.2 The models

Logistic Regression (LR)

One linear layer, one output. Simple, interpretable, the field-standard fairness baseline.

MLP (Multi-Layer Perceptron), 64–64

Input $\rightarrow 64 \rightarrow \text{ReLU} \rightarrow 64 \rightarrow \text{ReLU} \rightarrow 1$. A small neural network. Its role is to answer RQ2’s “does this hold for a nonlinear model too?”

ReLU

Activation function, $\max(0, x)$, gives the network nonlinearity.

Adam

The optimiser. Learning rate 0.01, weight decay 10^{-4} .

Weight decay (L2 regularisation)

Shrinks weights toward zero to prevent overfitting.

Early stopping

Stop training when validation loss stops improving. Patience = 20 epochs, and it *deep-copies* the best weights rather than holding a live reference — that was a real bug in the legacy code.

Training detail worth stating explicitly

The training loop does **one optimiser step per epoch on the full dataset** — full-batch gradient descent, not mini-batch SGD. With `num_epochs=300` that is at most 300 gradient steps. This is a deliberate consequence of the loss design (SoftDP needs all groups present in the batch to compute a group gap), and it is fine for tabular data of this size. But say it explicitly — reviewers will otherwise assume mini-batching.

2.3 The knobs: α and β

The whole project lives in this equation:

$$\mathcal{L} = \alpha \cdot \text{BCE} + (1 - \alpha) \left[\beta \cdot \text{SoftDP} + (1 - \beta) \cdot \text{SoftGE} \right] \quad (1)$$

Knob	Value	Meaning
α	1	Pure BCE, no fairness term at all. This is the base-line .
	0	Fairness only, accuracy ignored $\rightarrow$ degenerate models.
	swept	{0, 0.005, 0.05, 0.25, 0.5, 0.75, 1}
β	1	All fairness weight on group fairness (SoftDP).
	0	All fairness weight on individual fairness (SoftGE).
	0.5	Equal blend. Same seven swept values.

The log-ish spacing (0, 0.005, 0.05, 0.25, ...) resolves the region near zero, where even small fairness weights already change behaviour.

Why β is the important one

Sweeping β at fixed α holds the *total* fairness pressure constant while redistributing it between the two notions. That is exactly the comparison needed to isolate a trade-off between them — any change you see is attributable to *which* fairness you asked for, not *how much*.

2.4 Surrogates — the key technical trick

The real fairness metrics require **hard 0/1 predictions**, which come from a threshold, which is a step function, which has **zero gradient everywhere**. You cannot backpropagate through it. So the metrics as normally computed cannot be used as a training loss.

The fix: build **differentiable surrogates** that operate on the raw probabilities p instead of thresholded predictions.

2.4.1 SoftDP — soft demographic parity

$$\text{SoftDP} = \max_g \bar{p}_g - \min_g \bar{p}_g \quad (2)$$

The largest gap in *mean predicted probability* between any two sensitive groups. As probabilities saturate toward 0/1 this converges to the hard DPD.

2.4.2 SoftGE — soft generalized entropy

$$b_i = 1 + p_i - y_i, \quad \text{SoftGE} = \frac{1}{n\gamma(\gamma-1)} \sum_i \left[\left(\frac{b_i}{\bar{b}} \right)^\gamma - 1 \right], \quad \gamma = 2 \quad (3)$$

The **benefit vector** b is the central idea. For each person, $b_i = 1 + \text{prediction} - \text{truth}$:

Situation	b_i	Interpretation
Correct prediction	1	you got what you deserved
False positive	2	you got <i>more</i> than you deserved
False negative	0	you got <i>less</i> than you deserved

Then measure the **inequality** of that benefit vector using an economics inequality index. If everyone gets $b = 1$, inequality is 0 — perfectly individually fair. The more the errors concentrate, the higher the index.

Worked Example 2.1: Computing the benefit vector and GE(2) by hand

Take eight people. True labels $y = [1, 0, 1, 0, 1, 0, 1, 0]$.

Case A — perfect predictions. $\hat{y} = [1, 0, 1, 0, 1, 0, 1, 0]$.

$$b = [1, 1, 1, 1, 1, 1, 1, 1], \quad \bar{b} = 1, \quad \text{GE}(2) = 0$$

Everyone receives identical benefit, so inequality is exactly zero.

Case B — one false positive (person 2 predicted 1, truth 0). $\hat{y} = [1, 1, 1, 0, 1, 0, 1, 0]$.

$$b = [1, 2, 1, 1, 1, 1, 1, 1], \quad \bar{b} = \frac{9}{8} = 1.125$$

$$\text{GE}(2) = \frac{1}{2} \left[\frac{1}{8} \sum_i \left(\frac{b_i}{1.125} \right)^2 - 1 \right] = \frac{1}{2} \left[\frac{7(0.7901) + 3.1605}{8} - 1 \right] = \frac{1.0864 - 1}{2} = \mathbf{0.0432}$$

Case C — one false negative (person 1 predicted 0, truth 1). $\hat{y} = [0, 0, 1, 0, 1, 0, 1, 0]$.

$$b = [0, 1, 1, 1, 1, 1, 1, 1], \quad \bar{b} = \frac{7}{8} = 0.875$$

$$\text{GE}(2) = \frac{1}{2} \left[\frac{7(1.30612) + 0}{8} - 1 \right] = \frac{1.14286 - 1}{2} = \mathbf{0.0714}$$

What this teaches: one error of *either* kind creates inequality, but the false negative scores higher (0.0714 vs 0.0432). Denying someone what they were owed registers as more unequal than giving someone a windfall. All three values were verified against the project's actual `IndividualFairness.py`.

2.4.3 Surrogate validation

Because we train on the soft version but report the hard version, we must show they track each other. Pearson correlations across all sweep points and seeds:

Dataset	$r(\text{SoftDP}, \text{DPD}_2)$	$r(\text{SoftGE}, \text{GE}(2))$
COMPAS	0.794	0.832
German	0.625	0.919
Adult	0.907	0.923

German’s SoftDP correlation (0.625) is the weak one — expected, since German is tiny ($n = 1000$, test split = 200) so the hard DPD is very noisy.

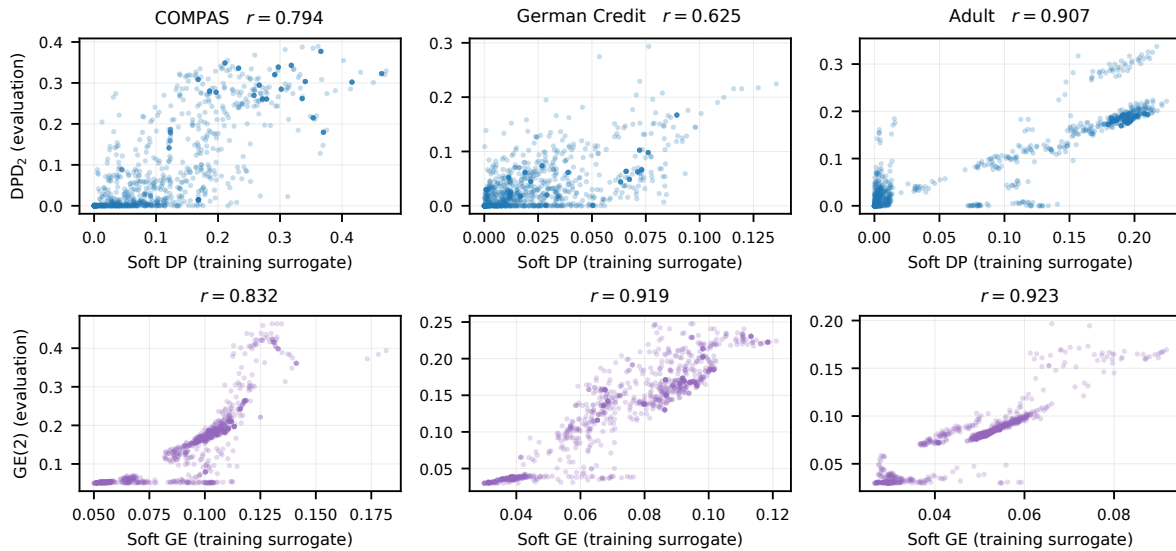


Figure 1: Each training surrogate plotted against the hard evaluation metric it stands in for, across all 2,940 models. Strong positive correlation is what licenses training on the soft version and reporting the hard one.

2.5 The baseline methods

Fairness interventions are conventionally sorted into three families. This project covers all three.

Family	Method	What it does
Pre-processing	Reweighting	Give each training sample weight $w(s, y) = P(s)P(y)/P(s, y)$, making label and sensitive attribute independent in the <i>weighted</i> training distribution.
In-processing	ExponentiatedGradient	Reformulates constrained fair classification as a two-player game; returns a randomised classifier. Run with Demographic Parity and with Equalized Odds.
Post-processing	ThresholdOptimizer	Keeps the trained model, picks a <i>different decision threshold per group</i> to satisfy the constraint.
(none)	Random Forest, XGBoost	No fairness intervention. These show the accuracy ceiling.

2.6 Experimental-protocol terms

Seed

A random number controlling the train/val/test split *and* weight initialisation *and* stochastic

baselines. Ten seeds (0–9) means ten independent repeats; we report mean $\pm$ std over them. This is what turns single numbers into results with error bars.

Train / validation / test split

60/20/20, re-drawn per seed. Train fits the model; validation selects hyperparameters and triggers early stopping; test is touched **only** for final reporting.

StandardScaler

Rescale numeric features to mean 0, std 1. Necessary because gradient descent behaves badly when features have wildly different scales (`age` $\approx$ 40, `capital-gain` $\approx$ 100,000).

OneHotEncoder(drop='first')

Turn a categorical column into binary indicator columns, dropping one to avoid perfect collinearity.

Utopia-point model selection

Among all $\alpha < 1$ sweep points, pick the one minimising distance to the ideal corner:

$$d = \sqrt{(1 - \text{val_acc})^2 + \text{val_DPD}_2^2}$$

Computed on **validation only**. This is a project-specific choice, not a field standard.

Degenerate-predictor guard

A model predicting “no” for everyone has $\text{DPD} = 0$, is perfectly “fair,” and is useless. Any model whose validation positive rate is $< 5\%$ or $> 95\%$ is excluded from selection.

Wilcoxon signed-rank test

A paired, non-parametric significance test. *Paired* because seed k for the fair model and seed k for the baseline share the same data split. *Non-parametric* because with ten samples you cannot credibly assume normality.

Holm correction

Testing four metrics at once gives four chances to find a fluke. Holm’s step-down procedure adjusts the p -values to control the family-wise error rate.

Pareto frontier

The set of configurations where you cannot improve accuracy without worsening DPD, or vice versa. The “efficient” trade-offs.

Worked Example 2.2: Why the degeneracy guard is not cosmetic

On COMPAS, seed 0, at $\alpha = 0$ (no accuracy term at all), the trained model records:

$$\text{accuracy} = 0.4526 \quad \text{validation positive rate} = 0.9951$$

The model predicts “will re-offend” for 99.5% of people. Its demographic parity difference is near zero, because it treats everyone identically — identically badly. Without the 5%/95% guard, model selection would rank this as an excellent fair model.

2.7 Abbreviations you will use constantly

Abbr.	Meaning	Abbr.	Meaning
LR	Logistic Regression	DPD	Demographic Parity Difference
MLP	Multi-Layer Perceptron	EOD	Equalized Odds Difference
BCE	Binary Cross-Entropy	GE	Generalized Entropy (index)
DP	Demographic Parity	RF	Random Forest
EO	Equalized Odds	XGB	XGBoost
ExpGrad	ExponentiatedGradient	TPR/FPR	True/False Positive Rate

3 Evaluation Metrics and Their Interpretation

Six metrics are computed on every model. This section explains each one, how to read its value, and — critically — whether any standard threshold actually exists.

Read this before quoting any threshold

There is **no universally accepted “good” value** for DPD, EOD, or GE. This guide distinguishes three levels of authority:

- **Established standard** — a formal, citable rule. Almost nothing in fairness metrics qualifies.
- **Common rule of thumb** — widely used in practice but with no formal backing. Label it as such in the paper.
- **Project-specific interpretation** — our own reading of our own numbers.

3.1 Accuracy

What: fraction of correct predictions. **Direction:** higher is better. **Range:** 0 to 1.

Reference point: the majority-class rate. Base rates in this project are COMPAS 0.455 (majority-class ≈ 0.545), German 0.300 (≈ 0.70), Adult 0.248 (≈ 0.752). A model must beat that to be doing anything at all.

Standard? Accuracy itself is universal; there is **no universal “good” accuracy**. What exists is *literature-typical* performance per dataset — COMPAS models land at ≈ 0.66 – 0.70 , Adult ≈ 0.84 – 0.87 , German ≈ 0.72 – 0.77 .

3.2 DPD — Demographic Parity Difference

Stands for: Demographic Parity Difference (also: statistical parity difference).

Measures: the largest gap in **selection rate** (fraction predicted positive) between sensitive groups:

$$\text{DPD} = \max_g P(\hat{y} = 1 \mid S = g) - \min_g P(\hat{y} = 1 \mid S = g) \quad (4)$$

It ignores the true labels entirely — it only asks who gets flagged, not who deserved it.

Direction: lower is better. **Range:** 0 to 1. Zero means every group is flagged at an identical rate.

Is there a standard threshold? No.

What people *do* cite:

- **The four-fifths rule / 80% rule** (US EEOC, *Uniform Guidelines on Employee Selection Procedures*, 1978) — a legal rule of thumb for adverse impact. But it is a **ratio** test (min rate/max rate ≥ 0.8), **not a difference test**. It does not translate to a DPD threshold without knowing the base selection rate. Cite it for context; do *not* claim “DPD < 0.2 passes the 80% rule” — that is only true if the maximum selection rate happens to be 1.0.
- **Common informal practice:** DPD < 0.05 is treated as “close to parity,” 0.05–0.10 as modest disparity, > 0.10 as substantial. This is a *convention*, not a standard.

3.2.1 The two variants in this project

- Demographic Parity Difference — computed over **all** groups.

- DPD (Largest 2 Groups), written DPD_2 — restricted to the two largest groups. On COMPAS that is African-American ($n = 3175$) vs Caucasian ($n = 2103$), ProPublica’s focal comparison. On German and Adult (two groups each) the two variants are **identical by construction**.

Worked Example 3.1: Why the largest-2 variant exists

COMPAS has six race groups and the small ones are tiny: Asian $n = 31$, Native American $n = 11$ across the *whole* dataset — roughly 6 and 2 people in a 1,235-row test split. A max – min over all groups is then pinned by a handful of individuals and swings wildly. The evidence is in our own results:

COMPAS, LR baseline	mean	std
DPD (all six groups)	0.608	± 0.211
DPD_2 (two largest)	0.304	± 0.049

The standard deviation is over four times larger for the all-groups version. That is why DPD_2 is the primary metric — and this table *is* the justification to put in the paper.

3.3 EOD — Equalized Odds Difference

Measures: whether the model’s *error rates* are equal across groups. Computed as the larger of {max–min true-positive-rate gap, max–min false-positive-rate gap}. Unlike DPD it *does* use the true labels — it asks whether the model is equally accurate for each group, not whether it selects them equally often.

Direction: lower is better. **Range:** 0 to 1. **Standard threshold?** No — same situation as DPD.

The relationship that drives our biggest result

DPD and EOD are **mathematically incompatible** whenever base rates differ across groups. Our data has exactly that: Adult female base rate 0.114 vs male 0.312; COMPAS African-American 0.523 vs Caucasian 0.391. So **driving DPD to zero must push EOD up**. Our Adult results show this precisely.

COMPAS EOD is unreliable in the current code

EOD is computed over **all** groups, without the largest-2 restriction that was applied to DPD. So COMPAS EOD inherits exactly the tiny-group noise problem — which is why every COMPAS EOD is ≈ 0.64 – 0.83 with $\text{std} \approx 0.15$ – 0.22 , and why *no* COMPAS EOD comparison reaches significance. Either add a largest-2 EOD, or explicitly caveat the column. Do not report it as a clean finding.

3.4 GE(2) — Generalized Entropy Index (labelled “Theil” in the code)

What it is: inequality of the benefit vector $b = 1 + \hat{y} - y$, measured with the generalized entropy index of order 2:

$$\text{GE}(2) = \frac{1}{2n} \sum_i \left[\left(\frac{b_i}{\bar{b}} \right)^2 - 1 \right] = \frac{\text{Var}(b)}{2\bar{b}^2} \quad (5)$$

Direction: lower is better. **Range:** 0 to ∞ in principle; 0 means perfectly equal benefit (a perfect classifier, since every $b = 1$).

Naming error to fix before submitting

`IndividualFairness.py` defines `theil_index` as AIF360's `generalized_entropy_error(..., alpha=2)`. That is **backwards relative to the standard definition**. AIF360's own docstring states: a value of 0 is the mean log deviation, 1 is the **Theil index**, and 2 is half the squared coefficient of variation.

So everything the project calls “Theil Index” is really **GE($\alpha = 2$)**. **No number is wrong** — GE(2) is a perfectly standard inequality index, and crucially it is the *same* index the SoftGE surrogate optimises (also $\alpha = 2$), so training and evaluation are consistent. But the **label** must change, or a reviewer familiar with Speicher et al. will flag it immediately. The paper produced alongside this guide already uses **GE(2)** throughout.

What it actually captures on hard predictions: with binary predictions, $b \in \{0, 1, 2\}$ only. So GE(2) here is a **function of the false-positive and false-negative counts alone**. It measures how much error there is and how it splits between the two error types — not, strictly, whether *similar individuals* got similar treatment. This is the standard Speicher et al. operationalisation, so it is defensible, but be precise: it is an *inequality-of-outcomes* measure, not a Lipschitz-style “similar individuals treated similarly” measure.

Standard threshold? None whatsoever. GE has no absolute interpretation. It is meaningful **only comparatively** — model A versus model B on the same dataset. Say this plainly in the paper rather than implying a range.

3.5 Gini Coefficient

What it is supposed to be: the classic economics inequality measure, from the Lorenz curve. 0 = perfect equality, 1 = maximum inequality.

This metric carries no fairness information

`IndividualFairness.py` applies the Lorenz/Gini construction to `y_pred` directly. Since `y_pred` is a vector of 0s and 1s, this reduces **exactly** to

$$\text{Gini} = 1 - \text{positive prediction rate}$$

Verified numerically to four decimal places on three random vectors (positive rate 0.3270 $\rightarrow$ Gini 0.6730; 0.2045 $\rightarrow$ 0.7955; 0.4700 $\rightarrow$ 0.5300), and consistent with the result tables (German LR baseline Gini 0.869 implies a positive rate of 0.131 against a base rate of 0.30 — plausible under-prediction on an imbalanced dataset).

Recommendation: either **drop it** from the paper, or **relabel it** as “selection rate (reported as 1–Gini)” — which is genuinely useful, since it is exactly the quantity the degeneracy guard watches. What you must not do is present it as an independent individual-fairness metric.

3.6 Soft DP / Soft GE

Logged per sweep point, computed on *test probabilities*. These are not evaluation metrics — they exist solely for the surrogate-validation scatter (Figure 1). Keep them in supplementary material.

3.7 Summary table

Metric	Meaning (plain)	Dir.	Range	Standard?	How we interpret our result
Accuracy	% correct	↑	0–1	No. Compare to majority-class rate and dataset literature	COMPAS ≈ 0.68 , Adult ≈ 0.85 , German ≈ 0.73 — all literature-typical
DPD (all groups)	Max–min selection rate over all groups	↓	0–1	No. 80% rule is a <i>ratio</i> test, not a difference test	On COMPAS dominated by tiny-group noise — report as secondary only
DPD ₂	Same, restricted to the two biggest groups	↓	0–1	No. “ <0.05 = near parity” is convention, not standard	Primary group-fairness metric. Stable across seeds
EOD	Max gap in TPR <i>or</i> FPR across groups	↓	0–1	No. “ <0.1 ” is convention only	Mathematically opposed to DPD when base rates differ. COMPAS values unreliable
GE(2)	Inequality of benefit $b = 1 + \hat{y} - y$	↓	0– ∞	None. Comparative only	Primary individual-fairness metric. Only meaningful vs another model on the same data
Gini	<i>As implemented:</i> exactly 1–selection rate	—	0–1	n/a	Not an independent fairness metric. Drop or relabel
Soft DP / Soft GE	Differentiable training surrogates	↓	—	n/a	Supplementary: validates soft $\approx$ hard ($r = 0.63\text{--}0.92$)

↑ = higher is better, ↓ = lower is better.

4 Methodology, Step by Step

4.1 Step 1 — Dataset selection and loading

Three datasets, chosen because together they are the field-standard trio for tabular fairness work. Actual figures, read from the loaders:

	COMPAS	German Credit	Adult
Samples n	6,172	1,000	45,222
Target ($y = 1$)	re-offended	bad credit	income $> \$50K$
Sensitive attribute	race	Sex	sex
Number of groups	6	2	2
Overall base rate	0.455	0.300	0.248
Encoded features	12	21	≈ 79
<i>Group sizes and per-group base rates</i>			
Largest group	Afr.-Am. 3,175 (0.523)	male 690 (0.277)	Male 30,527 (0.312)
2nd largest	Caucasian 2,103 (0.391)	female 310 (0.352)	Female 14,695 (0.114)
Others	Hispanic 509, Other 343, Asian 31, Nat.Am. 11	—	—

Why these three: they span three domains (criminal justice, credit, income), two sensitive attributes (race, sex), and three orders of magnitude of sample size — exactly what RQ2 needs to test consistency.

Why the differing base rates matter: they are the *reason* DPD and EOD conflict. Adult’s female-vs-male base rate gap (0.114 vs 0.312, a factor of 2.7) is the mechanical driver of our most striking result.

4.2 Step 2 — The COMPAS leakage fix

We apply ProPublica’s standard row filters:

- `days_b_screening_arrest` within ± 30 days
- `is_recid` $\neq -1$
- `c_charge_degree` $\neq 0$
- `score_text` $\neq$ N/A

Features used: `age`, `sex`, `juv_fel_count`, `juv_misd_count`, `juv_other_count`, `priors_count`, `c_charge_degree`, `race`.

This is a genuine methodological contribution

An earlier version of this project included a feature `duration = end - start` that correlated **-0.78** with the target. That correlation is **mechanical, not predictive**: re-offending truncates the observation window, so short duration $\Rightarrow$ re-offended, almost by definition. That is **label leakage** — the model was reading the answer off the feature. Removing it drops COMPAS accuracy to the literature-typical ≈ 0.67 – 0.68 . **This is a correction, not a regression.** Frame it as a soundness contribution in the paper.

4.3 Step 3 — Preprocessing

Per seed:

1. Split 60/20/20 into train/validation/test. (Note: splits are *not* stratified — worth a sentence in the paper.)
2. Extract the sensitive column per split; build integer group codes with **one shared category mapping across all splits** so group IDs mean the same thing everywhere. This matters — the torch loss indexes by these codes.
3. Fit the ColumnTransformer **on train only**, apply to val/test — correct, no leakage:
 - numeric $\rightarrow$ StandardScaler
 - categorical $\rightarrow$ OneHotEncoder(drop='first', handle_unknown='ignore')
4. Convert to torch tensors.

There is no feature-engineering step — say so

No interaction terms, no binning, no feature selection. Features are used as published. This is *appropriate* — the paper is about the loss function, and inventing features would confound the comparison. But state it explicitly rather than leaving a gap where a “feature engineering” section would be.

4.4 Step 4 — The composite loss

Equation (1), with SoftDP and SoftGE as defined earlier. Two bugs from the legacy code are fixed here, and both belong in the paper’s reproducibility discussion:

1. **Gradient flow.** The original computed fairness terms via fairlearn/aif360 on the detached array

```
y_pred.round().detach().numpy()
```

The `.detach()` severs the autograd graph — **only BCE ever produced gradients**, so α and β had almost no effect on the learned model. That is why the legacy heatmaps were flat in β . Every term is now computed in torch on raw probabilities.

2. **Double sigmoid.** The old model applied sigmoid in `forward` and the loss applied it again. Now models return logits and the loss uses `binary_cross_entropy_with_logits`.

Worked Example 4.1: Putting a number through the composite loss

Suppose at some point in training, on the training batch:

$$\text{BCE} = 0.62, \quad \text{SoftDP} = 0.18, \quad \text{SoftGE} = 0.09$$

Compare two settings of the dial at the same fairness pressure $\alpha = 0.75$ (so $1 - \alpha = 0.25$):

$\beta = 1$ (**group-only**):

$$\mathcal{L} = 0.75(0.62) + 0.25[1(0.18) + 0(0.09)] = 0.465 + 0.045 = \mathbf{0.510}$$

$\beta = 0$ (**individual-only**):

$$\mathcal{L} = 0.75(0.62) + 0.25[0(0.18) + 1(0.09)] = 0.465 + 0.0225 = \mathbf{0.4875}$$

$\beta = 0.5$ (**blend**):

$$\mathcal{L} = 0.465 + 0.25[0.5(0.18) + 0.5(0.09)] = 0.465 + 0.03375 = \mathbf{0.4988}$$

Same accuracy term in all three; only the *composition* of the fairness penalty changes. Gradient descent therefore pushes the model in a different direction for each β — which is the entire mechanism the study exploits.

4.5 Step 5 — Training

Adam (lr 0.01, weight decay 10^{-4}), full-batch, up to 300 epochs, early stopping on **validation composite loss** with patience 20, best weights restored from a deep copy.

Note that early stopping monitors the *composite* loss at that (α, β) — not plain BCE. This is the right choice (you select the best model for the objective you are actually optimising), but it means the stopping criterion shifts as you move across the grid. Mention it.

4.6 Step 6 — The sweep

For each of $7\alpha \times 7\beta = 49$ combinations, per architecture (2), per dataset (3), per seed (10):

- Set `torch.manual_seed(seed)` and `np.random.seed(seed)` before instantiating the model $\Rightarrow$ identical initialisation across grid points within a seed, so differences are attributable to (α, β) , not initialisation luck.
- Train (or load a cached checkpoint).
- Predict on test, threshold at 0.5, compute all six metrics.
- Also record validation accuracy, validation DPD_2 , validation positive rate (for selection and the degeneracy guard), soft metrics, and training time.

Total: $49 \times 2 \times 3 \times 10 = 2,940$ **trained models.**

4.7 Step 7 — Model selection

Validation-only, in three steps:

1. Keep $\alpha < 1$ (must have *some* fairness weight).

2. Keep validation positive rate $\in [0.05, 0.95]$ (degeneracy guard).

3. Minimise $\sqrt{(1 - \text{val_acc})^2 + \text{val_DPD}_2^2}$.

4. Test metrics reported only, **never used for selection**.

Which (α, β) actually got picked, across ten seeds — this is itself a result:

Dataset	Model	Distinct configs / 10 seeds	Mean β
COMPAS	Fair LR	5	0.63
COMPAS	Fair MLP	5	0.60
German	Fair LR	8	0.19
German	Fair MLP	10	0.41
Adult	Fair LR	4	0.58
Adult	Fair MLP	3	0.65

German is unstable: the MLP picks a *different* configuration in every single seed. That instability is a finding, analysed in Section 6.4.

4.8 Step 8 — Baselines and Step 9 — Aggregation

All baselines are fit on the same seed’s training split and evaluated on the same test split. Then the ten per-seed CSVs are concatenated, and for each (fair model, baseline) pair we run a **paired Wilcoxon signed-rank test** across seeds per metric, **Holm-corrected** across the metric family {Accuracy, DPD_2 , EOD, $\text{GE}(2)$ }.

A statistical detail to get right

With $n = 10$ seeds, the smallest two-sided p -value a Wilcoxon test can produce is $2^{-9} \approx 0.00195$. Report it as $p = 0.002$. Do **not** write “ $p < 0.001$ ” — that is not attainable at this sample size, and it means every single seed moved in the same direction.

4.9 Libraries used

Library	Used for
PyTorch	models + the differentiable composite loss
scikit-learn	preprocessing, splits, LR/RF, accuracy
fairlearn	DPD, EOD, ExponentiatedGradient, ThresholdOptimizer
AIF360	generalized entropy index
xgboost	XGBoost accuracy reference
scipy	Wilcoxon signed-rank test
pandas / numpy	data handling
matplotlib / seaborn	figures

5 Methodology Flowchart

Figure 2 shows the complete pipeline as actually implemented. Note that the generic template (data → preprocessing → feature engineering → model → evaluation) is modified in two ways for this project: there is **no feature engineering stage** (deliberate — features are used as published), and the model stage **branches** into the sweep and the baselines before rejoining at evaluation.

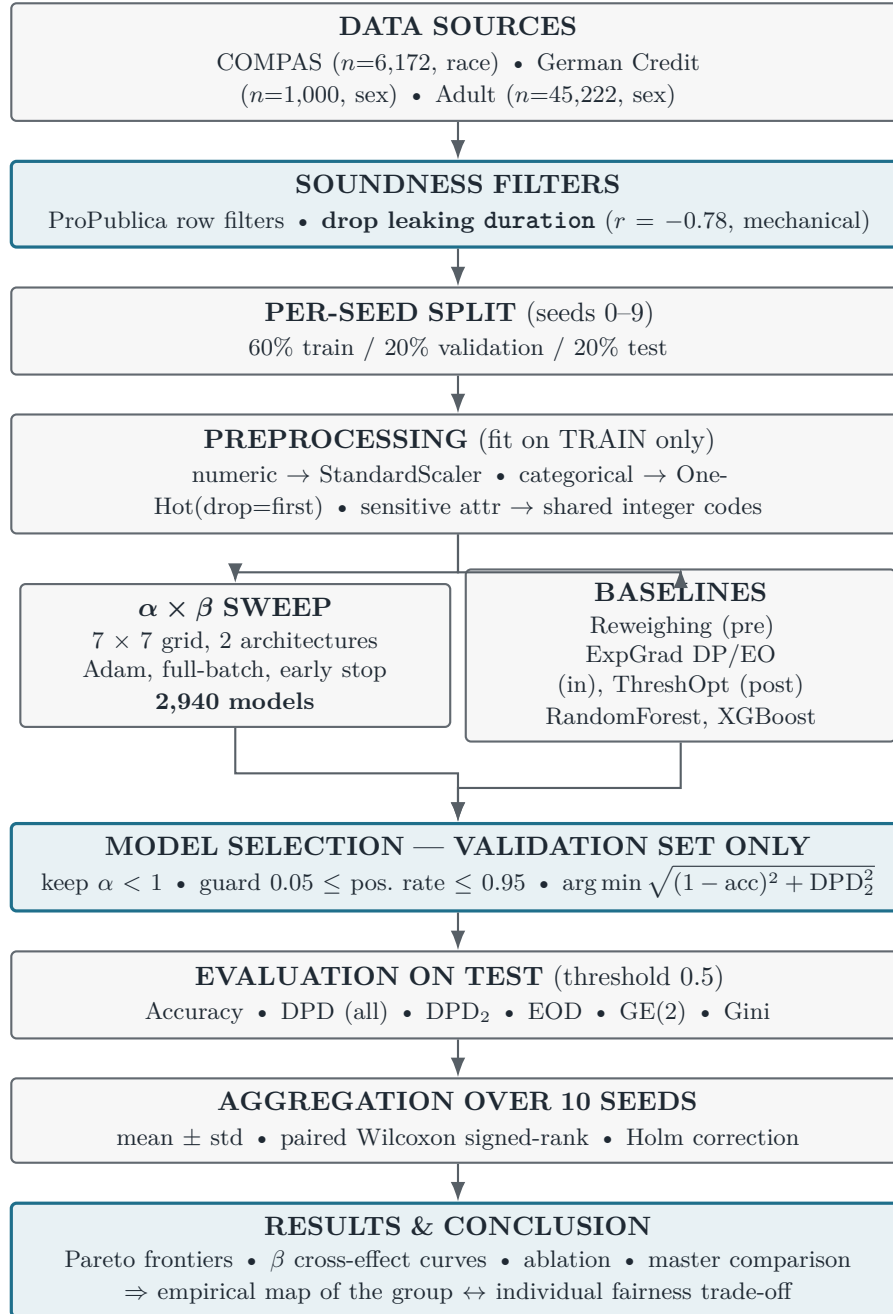


Figure 2: The complete pipeline. Highlighted boxes are the three steps that most affect the validity of the conclusions.

6 Understanding Our Results

6.1 Headline: the β ablation — this is the paper

This is the cleanest evidence for RQ1. It holds α fixed and moves β between the extremes, so the *amount* of fairness pressure is constant and only its *composition* changes.

COMPAS, Logistic Regression ($\alpha = 0.75$)

Variant	Accuracy	DPD ₂ ↓	GE(2) ↓
No fairness ($\alpha=1$)	0.676 ± 0.012	0.304 ± 0.049	0.191 ± 0.012
SoftGE only ($\beta=0$)	0.674 ± 0.012	0.305 ± 0.049	0.181 ± 0.010
SoftDP only ($\beta=1$)	0.650 ± 0.014	0.067 ± 0.053	0.217 ± 0.021
Blend ($\beta=0.5$)	0.660 ± 0.021	0.092 ± 0.065	0.204 ± 0.026

Read it row by row — this is the whole story:

- Optimising **individual fairness alone** ($\beta = 0$): GE(2) improves 0.191 → 0.181 (−5%). DPD₂ is **completely unmoved**: 0.304 → 0.305. It does nothing for group fairness.
- Optimising **group fairness alone** ($\beta = 1$): DPD₂ collapses 0.304 → 0.067 (−78%). But GE(2) gets **worse**: 0.191 → 0.217 (+14%) — *worse than doing nothing at all*.
- The blend sits in between on both.

The same pattern appears on Adult, and it is statistically significant. Full ablation with Holm-corrected significance:

Dataset	Variant	Accuracy	DPD ₂	EOD	GE(2)
COMPAS	No fairness	0.676	0.304	0.789	0.191
	SoftGE only ($\beta=0$)	0.674	0.305 NS	0.811*	0.181**
	SoftDP only ($\beta=1$)	0.650**	0.067**	0.681 NS	0.217**
German	No fairness	0.730	0.046	0.090	0.175
	SoftGE only ($\beta=0$)	0.730 NS	0.050 NS	0.106 NS	0.170 NS
	SoftDP only ($\beta=1$)	0.731 NS	0.029 NS	0.103 NS	0.173 NS
Adult	No fairness	0.845	0.184	0.107	0.084
	SoftGE only ($\beta=0$)	0.845 NS	0.189**	0.113*	0.083*
	SoftDP only ($\beta=1$)	0.833**	0.040**	0.246**	0.094**

Logistic regression. ** $p_{\text{Holm}} < 0.01$, * $p_{\text{Holm}} < 0.05$, NS = not significant.

The result, stated precisely: a double dissociation

1. Group fairness ← group objective: **strong** effect (−78% DPD, $p_{\text{Holm}} = 0.008$, consistent across datasets and architectures).
2. Individual fairness ← individual objective: **weak but real and significant** effect (−5% GE, $p_{\text{Holm}} \leq 0.012$).
3. **The cross-effects are negative in both directions.** Optimising group fairness significantly *harms* individual fairness. Optimising individual fairness leaves group fairness unchanged (COMPAS) or significantly *worse* (Adult).

Is this good? Yes — it is an excellent result, because it is clean, replicated, and significant in both directions.

Why the asymmetry (effect 1 strong, effect 2 weak)? This is itself a finding, not a flaw. SoftDP directly manipulates a quantity the model controls freely — the group-wise mean of predicted probabilities — and can shift it by moving the decision surface, which costs

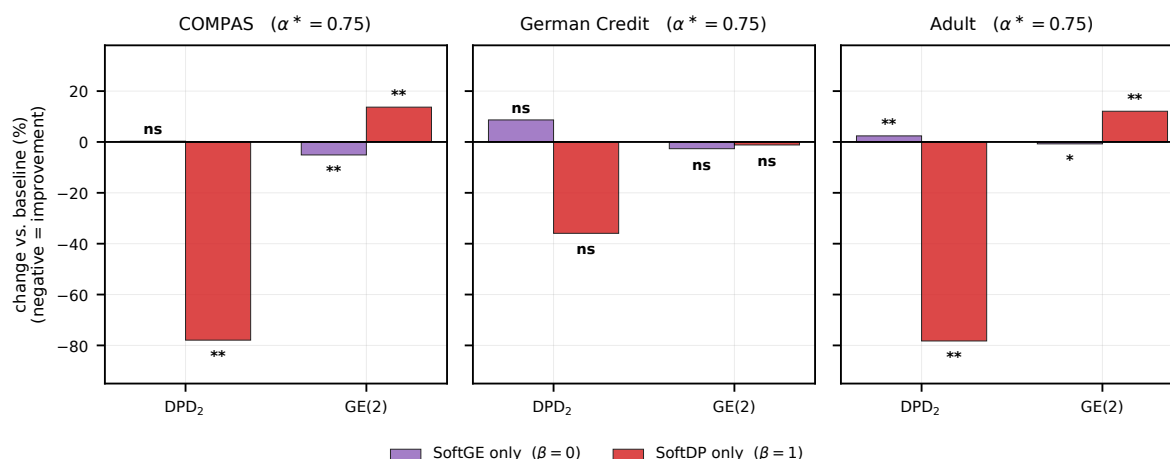


Figure 3: The double dissociation as percentage change from baseline. Purple ($\beta = 0$, individual-only) lowers $GE(2)$ but not DPD . Red ($\beta = 1$, group-only) crushes DPD but raises $GE(2)$. German shows no significant movement in either direction.

little accuracy. SoftGE is bounded by irreducible prediction error: benefit is equal only when everyone is predicted equally correctly, so you *cannot* equalise benefit without actually being more accurate. On COMPAS, where the accuracy ceiling is about 0.68, there is very little room.

6.2 The β cross-effect curves

6.3 Fair model versus baseline (the selected models)

Holm-corrected p -values for the selected fair model against its matched baseline:

Dataset	Arch	Accuracy	DPD_2	EOD	$GE(2)$
COMPAS	LR	0.008	0.008	0.232	0.027
COMPAS	MLP	0.018	0.008	0.863	0.863
German	LR	1.000	1.000	1.000	0.422
German	MLP	1.000	1.000	1.000	1.000
Adult	LR	0.008	0.008	0.008	0.008
Adult	MLP	0.008	0.008	0.008	0.008

6.3.1 COMPAS — good result

DPD_2 drops $0.304 \rightarrow 0.061$ (LR), an **80% reduction**, at a cost of 1.5 accuracy points ($0.676 \rightarrow 0.661$). Both significant. $GE(2)$ worsens $0.191 \rightarrow 0.200$, also significant. **Verdict: good.** A five-fold fairness improvement for 1.5 points of accuracy is a favourable exchange rate by the standards of this literature, and the individual-fairness cost is measured rather than hidden.

6.3.2 Adult — the strongest and most interpretable result

DPD_2 $0.184 \rightarrow 0.021$ (**-89%**), accuracy $0.845 \rightarrow 0.830$ (-1.5 points), EOD $0.107 \rightarrow$ **0.279** ($2.6\times$ worse), $GE(2)$ $0.084 \rightarrow 0.095$. All four significant, all four moving in the same direction across all ten seeds.

Why Adult's "bad" EOD number is the best result in the paper

Adult's base rates are 0.114 (female) vs 0.312 (male) — a $2.7\times$ gap. Theory says you cannot have demographic parity and equalized odds simultaneously under unequal base

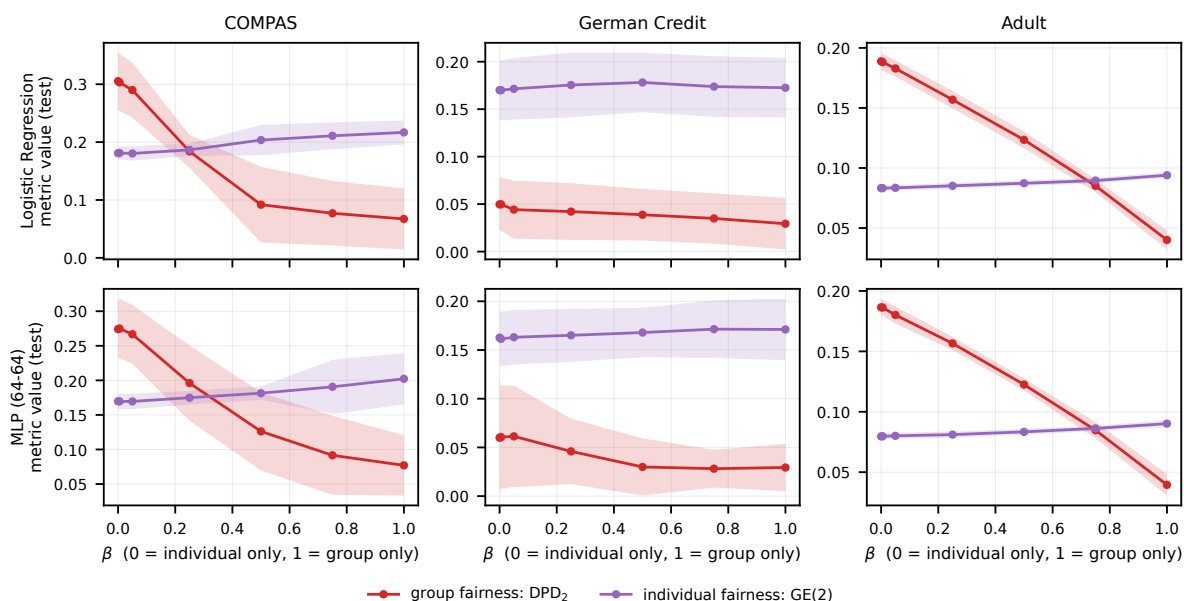


Figure 4: The trade-off seen continuously. As β moves from individual-only (0) to group-only (1), the group metric (red) falls and the individual metric (purple) rises. The **crossing curves** on COMPAS and Adult are the result; German is flat because there is almost no disparity to remove.

rates. Your experiment shows exactly that, at the predicted magnitude. This is a **textbook confirmation**, and you should present it as one. The exchange rate is roughly one point of DPD for 1.1 points of EOD.

6.4 The German Credit null result — and its real cause

On German, **nothing is significant** — not one metric, not one architecture. This must be reported as a null result, not buried. But the common-sense explanation (“the dataset is small”) is only half right, and the naive explanation (“the method doesn’t work here”) is **wrong**.

6.4.1 The decisive diagnostic

Model selection ranks configurations by *validation* DPD. So: does validation DPD predict test DPD?

Dataset	$r(\text{val DPD, test DPD})$	SNR	acc:DPD term ratio
COMPAS	0.94	1.68	1.1×
Adult	0.99	12.91	0.7×
German	0.08	0.45	34.9×

On German, validation DPD carries **essentially no information** about test DPD. Selection is choosing at random with respect to fairness. With a 200-row validation split containing ≈ 62 women, the standard error of a selection rate is $\approx \pm 0.04$ — the size of the entire signal. German is also the only dataset where the seed noise (SNR 0.45) is *larger* than the between-configuration signal.

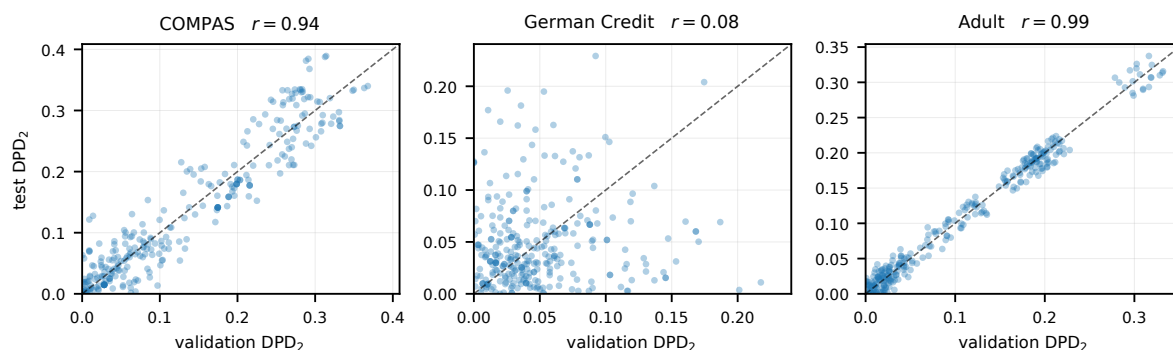


Figure 5: Validation versus test DPD_2 across sweep configurations. On COMPAS and Adult the points hug the diagonal — validation predicts test. On German they form a cloud.

6.4.2 A second, independent problem: the rule is scale-blind

The selection rule minimises $\sqrt{(1 - \text{acc})^2 + \text{DPD}^2}$. At German’s baseline, $(1 - \text{acc}) = 0.270$ but $\text{DPD} = 0.046$, so in squared terms the accuracy term dominates by **34.9×**. The rule collapses to “maximise accuracy” and is nearly blind to fairness — even with perfect, noise-free data. On COMPAS ($1.1\times$) and Adult ($0.7\times$) the terms are coincidentally balanced, which is why it works there.

Worked Example 6.1: The method *does* work on German — selection just can’t find it

Bypass selection entirely and fix the group-fairness extreme $\beta = 1$:

German, LogReg	Test DPD_2	Accuracy	Seeds improved
Baseline ($\alpha=1$)	0.046	0.730	—
$\alpha=0.5, \beta=1$	0.024	0.729	8/10 ($p = 0.020$)
What selection actually picked	0.047	0.726	—

A **48% disparity reduction at zero accuracy cost** was available. The utopia rule walked straight past it — across ten seeds it picked β values averaging 0.19, with *three seeds choosing* $\beta = 0$ (no group-fairness weight at all).

Caution: $\alpha = 0.5$ gives $p = 0.020$ but $\alpha = 0.75$ gives $p = 0.193$. Do not cherry-pick α — that would be exactly the test-set selection avoided everywhere else. Cite the *pre-specified* ablation contrast instead.

How to frame German in the paper

Not “the method fails on German.” Instead: *the selection procedure fails on German, and here is the diagnostic proving it.* This is a genuinely useful methodological finding — **validation-based fairness model selection has a sample-size floor** — because fairness metrics are differences of *conditional* rates, so their effective sample size is set by the smallest group, not by n .

6.5 Comparison against established baselines (RQ3)

Method	COMPAS		Adult	
	Accuracy	DPD ₂	Accuracy	DPD ₂
Reweighting	0.656	0.034	0.840	0.090
ExpGrad (DP)	0.658	0.035	0.830	0.013
ThresholdOptimizer	0.663	0.037	0.827	0.009
Ours (Fair LR)	0.661	0.061	0.830	0.021
Ours (Fair MLP)	0.669	0.082	0.836	0.026
XGBoost (no fairness)	0.655	0.228	0.868	0.181
Random Forest	0.624	0.164	0.843	0.193

Honest verdict: competitive, not superior

At matched accuracy, the dedicated group-fairness methods reach modestly lower DPD than the composite loss. **Do not oversell this — and you don’t need to.** Your contribution is the trade-off map, not a new state of the art.

The correct framing: *“the composite loss achieves group-fairness levels comparable to established methods, which establishes it as a credible instrument for probing the trade-off — and unlike those methods, it exposes a continuous, controllable interpolation between the two fairness notions.”* None of ExpGrad, ThresholdOptimizer, or Reweighting has a β -equivalent. That is your differentiator.

One more useful observation: **XGBoost on Adult** has the highest accuracy (0.868) and one of the largest disparities (0.181). The most accurate model is among the least fair — a clean one-sentence motivation for the whole paper.

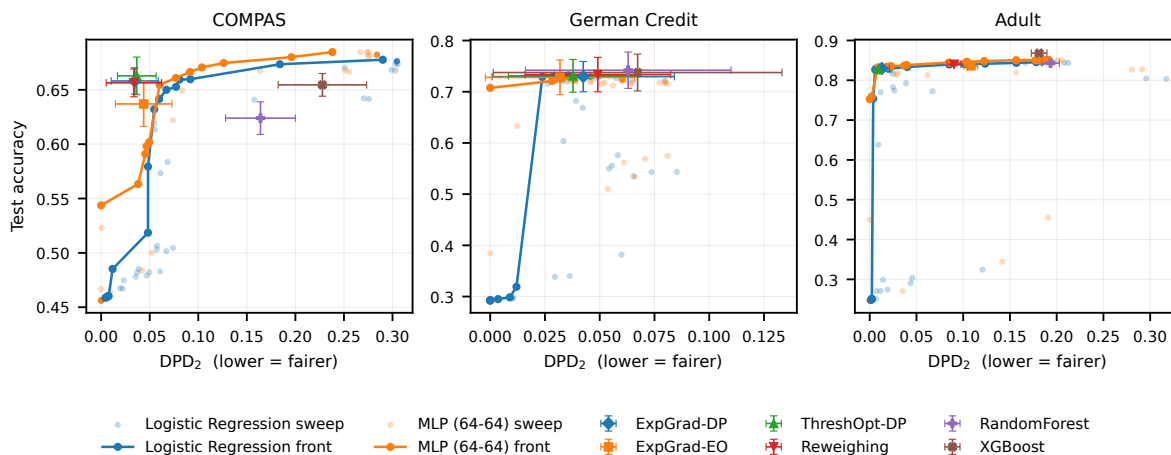


Figure 6: Accuracy versus DPD₂. Faint points are sweep configurations, solid lines the Pareto fronts, markers the baselines with error bars.

6.6 What can we reasonably conclude?

Supported by our data	NOT supported — do not claim
A group-fairness objective produces large, significant reductions in group metrics (up to -89% DPD) at ≈ 1.5 accuracy points, where a disparity exists.	That our method beats established baselines. It doesn't; it matches them.
That improvement carries a <i>measured</i> cost to individual fairness ($+5\text{--}14\%$ GE) and, on Adult, a severe cost to equalized odds ($2.3\text{--}3.3\times$).	Anything about individual fairness in the Dwork “similar individuals” sense — GE(2) is an inequality-of-outcomes measure.
An individual-fairness objective improves individual fairness weakly and group fairness not at all.	Any COMPAS EOD conclusion — that column is noise-dominated by 11-person groups.
The pattern holds across two architectures and two of three datasets; the third is explained, not contradictory.	Any conclusion resting on the Gini column — it is 1-selection rate.
The composite loss reaches group-fairness levels comparable to Reweighting / ExpGrad / ThresholdOptimizer.	Generalisation beyond tabular binary classification with a single sensitive attribute.

7 Research Paper Preparation

What belongs in each section, based on the methodology and results above.

7.1 Introduction

- Open with concrete stakes: COMPAS, credit, hiring. One sentence, no melodrama.
- Establish that “fairness” is not one thing — group vs individual, with a one-line intuition for each.
- State the gap: impossibility results tell us the definitions conflict *in principle*; far less work measures *how much*, with error bars, across datasets.
- State the contribution list explicitly (reviewers look for this): (1) a differentiable composite loss with a single interpolation parameter; (2) an empirical trade-off map over 2,940 models; (3) significance-tested comparison against all three mitigation families; (4) a documented label-leakage correction to a COMPAS feature set.
- Preview the headline: -89% DPD on Adult at -1.5 accuracy points, with EOD $2.6\times$ worse.

7.2 Related Work / Literature Review

Four blocks:

1. **Group fairness definitions & methods** — demographic parity, equalized odds (Hardt et al. 2016), Reweighting (Kamiran & Calders 2012), ExponentiatedGradient (Agarwal et al. 2018). Cite what you actually ran.
2. **Individual fairness** — Dwork et al. (2012) Lipschitz formulation; Speicher et al. (2018) benefit-vector formulation. **Be explicit that you use the Speicher operationalisation, not the Dwork one**, and say why (computable without a similarity metric).
3. **Impossibility & tension results** — Kleinberg et al. (2016); Chouldechova (2017); Barocas, Hardt & Narayanan. This is the theory your Adult result empirically confirms.
4. **COMPAS as a benchmark and its critiques** — ProPublica’s analysis, the rebuttals, and the feature-set/leakage literature. This is where your *duration* fix lives.

Position the gap in one sentence: prior work either proves conflict theoretically or optimises one notion at a time; you *interpolate continuously* and measure both.

7.3 Methodology

- The loss equation, with SoftDP and SoftGE derived and the benefit vector explained.
- **Why surrogates are necessary** — the thresholding step function has no gradient. Make this the motivating argument.
- Architectures, optimiser, full-batch training, early stopping.
- The validation-only utopia-distance selection rule and the 5%/95% degeneracy guard, both with justification.
- **Do not put results here.**

7.4 Experimental Setup

- Dataset table: n , target, sensitive attribute, group sizes, per-group base rate, encoded dimensionality.

- Preprocessing: 60/20/20 per seed, scaler + one-hot, fit on train only. Note splits are unstratified.
- The COMPAS filter list and the `duration` removal with its -0.78 correlation and mechanical explanation. Frame the accuracy drop as correctness.
- Grid: α, β value lists; total model count.
- Ten seeds control split + init + stochastic baselines.
- Baseline configurations with hyperparameters.
- Metric definitions **with the corrected GE(2) naming**, and the largest-2-groups DPD with its tiny-group justification (quote the Asian $n = 31$ / Native American $n = 11$ numbers — they make the argument for you).
- Statistical protocol, and note that $p = 1/512$ is the floor at $n = 10$.
- Reproducibility: library versions, CPU-only, checkpoint caching, exact commands.

7.5 Results

Order by strength, not by figure number:

1. **The β cross-effect and ablation.** Lead with this; it is the contribution.
2. **Fair vs baseline with significance.** COMPAS and Adult positive, German null.
3. **Adult's DPD/EOD inversion** — its own subsection, with the base rate numbers. Your theory-confirmation result.
4. **Pareto frontiers with baselines overlaid** (RQ3).
5. **German null result** — its own short subsection with the selection diagnostic. Reporting it plainly is a credibility asset.
6. **Supplementary** — surrogate validation, seed-mean heatmaps.

Every number as mean $\pm$ std over ten seeds. Every figure with error bars.

7.6 Discussion

- Interpret the **asymmetry**: why the group objective works strongly and the individual objective weakly (freely controllable quantity vs a floor set by irreducible error).
- Interpret the **Adult inversion** as the empirical face of the impossibility theorem, with the exchange rate quantified.
- **What β means for a practitioner**: not a hyperparameter to tune for performance, but a *policy dial* encoding which fairness notion the institution has committed to. This is the most quotable idea in the paper.
- **Why German shows nothing**, and what it implies about model selection at small n .
- Position vs baselines honestly.
- The uncomfortable normative point: Adult MLP at $\beta = 1$ gives DPD 0.009 — near-perfect demographic parity — while making error rates markedly *more* unequal. Fairer by one definition, less fair by another. Don't resolve it; that tension is the paper.

7.7 Limitations

Be thorough — it buys credibility, and every item is real and verifiable:

1. GE(2) is not Dwork-style individual fairness.
2. Gini as implemented equals 1–selection rate.
3. COMPAS EOD (and all-groups DPD) dominated by groups of 11–31 people.
4. German Credit underpowered at $n = 1000$; selection unstable.
5. Full-batch training, one gradient step per epoch, max 300 steps.
6. Single binary/categorical sensitive attribute; no intersectional analysis.
7. Fixed 0.5 threshold, unlike ThresholdOptimizer.
8. Utopia-distance selection implicitly weights 1 accuracy point = 1 DPD point, and is scale-sensitive.
9. Sensitive attribute included as a model input.
10. Adversarial debiasing not run.
11. Positive-class polarity inconsistent across datasets.
12. Tabular binary classification only.
13. Splits are unstratified.

7.8 Conclusion

Restate compactly: a differentiable composite loss lets β interpolate continuously between group and individual fairness; sweeping it across three datasets, two architectures and ten seeds produces an empirical map showing the group objective strongly reduces group disparity (-78% to -89% DPD) at a small accuracy cost (≈ 1.5 points) and a measurable cost to individual fairness and equalized odds; the individual objective helps individual fairness only weakly and group fairness not at all; the effect vanishes where the baseline is already near parity. Close on the practitioner-facing point: **β is a policy parameter, not a tuning knob** — the trade-off cannot be engineered away, only chosen.

7.9 Before you write: five fixes, ranked

#	Issue	Why it matters	Effort
1	Relabel “Theil Index” $\rightarrow$ GE($\alpha=2$)	Currently mislabelled against the standard definition; a knowledgeable reviewer will catch it. No numbers change.	Text only
2	Drop or relabel the Gini column	It is exactly 1–selection rate; presenting it as individual fairness is not defensible.	Text only
3	Add largest-2-groups EOD for COMPAS, or caveat it	COMPAS EOD is currently pinned by groups of 11 and 31 people.	≈ 10 lines + rerun
4	State explicitly: sensitive attribute is an input; training is full-batch; splits unstratified	All three are choices reviewers will ask about.	Text only
5	Frame German as a <i>reported null result</i> with its diagnostic	Reporting it plainly is a credibility asset and it supports RQ2.	Text only

Status

Fixes 1, 2, 4 and 5 have already been applied in the IEEE paper produced alongside this guide. Fix 3 is documented there as a stated limitation with the remedy named.

End of study guide.